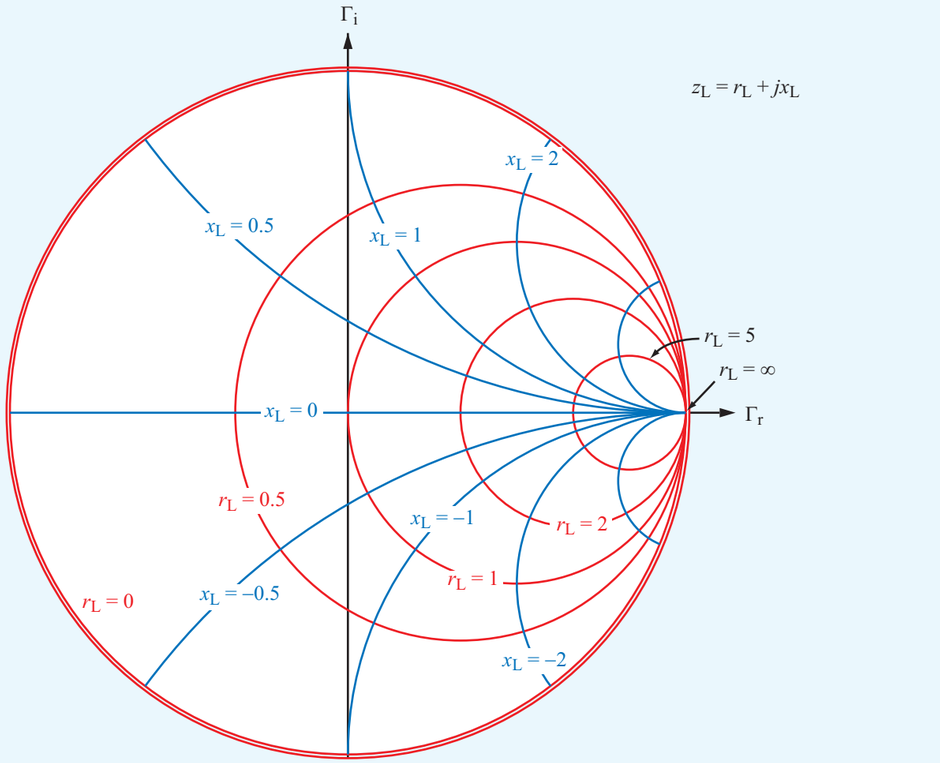


Figure 2-25 Families of r_L and x_L circles within the domain $|\Gamma| \le 1$. The r_L circles are concentric with the outermost circle corresponding to $r_L = 0$ and the innermost circle (of zero radius) corresponding to $r_L = \infty$. The parts of the x_L circles contained in the Smith chart are in the upper half of the chart if x_L is positive and in the lower half of the chart if x_L is negative.

which is the equation of a circle of radius $(1/x_L)$ centered at $(\Gamma_r, \Gamma_i) = (1, 1/x_L)$. The x_L circles in the Γ_r–Γ_i plane are quite different from those for constant r_L. To start with, the normalized reactance x_L may assume both positive and negative values, whereas the normalized resistance cannot be negative (negative resistances cannot be realized in passive circuits). Hence, Eq. (2.118) represents two families of circles: one for positive values of x_L and another for negative ones. Furthermore, as shown in **Fig. 2-25**, only part of a given circle falls within the bounds of the $|\Gamma| = 1$ unit circle.

The families of circles of the two parametric equations given by Eqs. (2.116) and (2.118) plotted for selected values of r_L and x_L constitute the Smith chart shown in **Fig. 2-26**. The Smith chart provides a graphical evaluation of Eqs. (2.115a and b) and their inverses.

► Each point in the Smith chart conveys the values of two interrelated quantities, namel z_L and Γ. The intersection of the r_L and x_L circles at that point defines $z_L = r_L + jx_L$. Simultaneously, the location of the point defines $|\Gamma|$ and θ_r. ◄

For example, point P in **Fig. 2-26** represents a normalized load impedance $z_L = 2 - j1$, which corresponds to a voltage reflection coefficient $\Gamma = 0.45\exp(-j26.6^\circ)$. The magnitude

86 CHAPTER 2 TRANSMISSION LINES

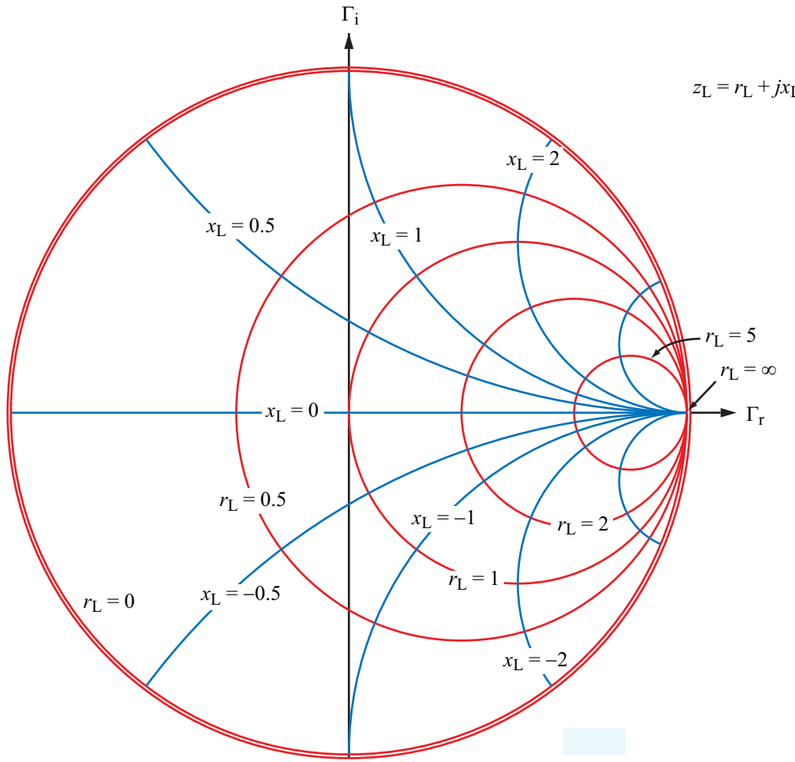


图 2-25 $|\Gamma| \le$ 域内的 r_L 和 x_L 圆族 1. r_L 圆与对应于 $r_L = 0$ 的最外圆和对应于 $r_L = \infty$ 的最内圆（半径为零）同心。如果 x_L 为正，则史密斯圆图中包含的 x_L 圆的部分位于图表的上半部分；如果 x_L 为负，则位于图表的下半部分。

这是以 $(\Gamma_r, \Gamma_i) = (1, 1/x_L)$ 为中心、半径为 $(1/x_L)$ 的圆方程。 Γ_r–Γ_i 平面中的 x_L 圆与常数 r_L 的圆有很大不同。首先，归一化电抗 x_L 可以取正值和负值，而归一化电阻不能为负值（负电阻不能在无源电路中实现）。因此，等式。 (2.118) 表示两个圆族：一个代表 x_L 的正值，另一个代表负值。此外，如图2-25所示，只有给定圆的一部分落在$|\Gamma| = 1$单位圆的边界内。

方程给出的两个参数方程的圆族。针对 r_L 和 x_L 的选定值绘制的 (2.116) 和 (2.118) 构成了图 2-26 所示的史密斯圆图。这

史密斯圆图提供了方程式的图形评估。 (2.115a 和 b）及其逆。

► 史密斯圆图中的每个点都表示两个相互关联的量的值，即 z_L 和 Γ。 r_L 和 x_L 圆在该点的交点定义了 $z_L = r_L + jx_L$。同时，该点的位置定义了$|\Gamma|$和θ_r。 ◄

例如，图2-26中的点P表示归一化负载阻抗$z_L = 2 - j1$，其对应于电压反射系数$\Gamma = 0.45\exp(-j26.6^\circ)$。震级

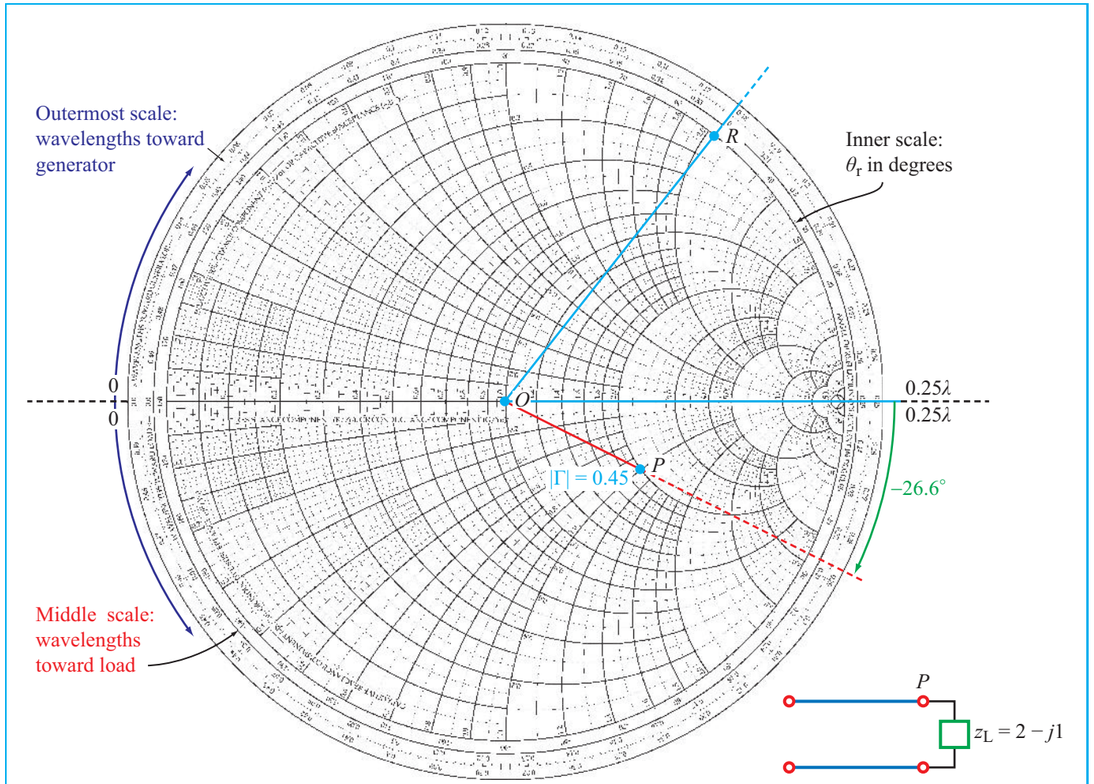
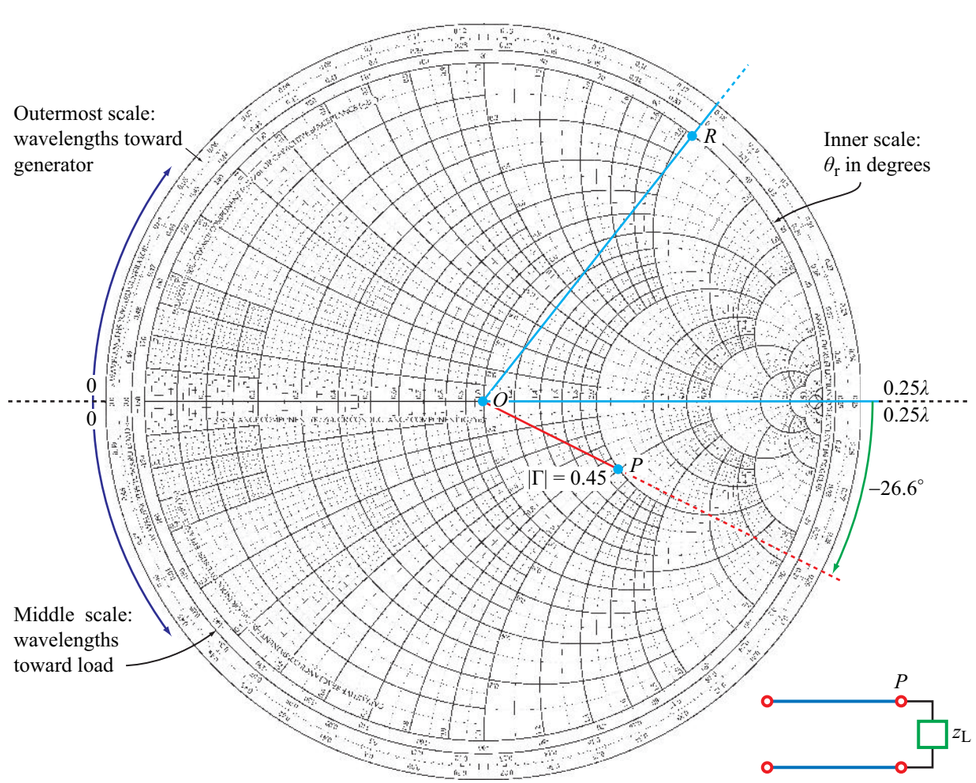


Figure 2-26 Point P represents a normalized load impedance $z_L = 2 - j1$. The reflection coefficient has a magnitude $|\Gamma| = \overline{OP}/\overline{OR} = 0.45$ and an angle $\theta_r = -26.6°$. Point R is an arbitrary point on the $r_L = 0$ circle (which also is the $|\Gamma| = 1$ circle).

$|\Gamma| = 0.45$ is obtained by dividing the length of the line between the center of the Smith chart and point P (designated $\overline{OP}$ in **Fig. 2-26**), by $\overline{OR}$, which is the length of the line between the center of the Smith chart and the edge of the unit circle (the radius of the unit circle corresponds to $|\Gamma| = 1$). The perimeter of the Smith chart contains three concentric scales. The innermost scale is labeled ***angle of reflection coefficient in degrees***. This is the scale for θ_r. As indicated in **Fig. 2-26**, $\theta_r = -26.6°$ (−0.46 rad) for point P. The meanings and uses of the other two scales are discussed next.

Exercise 2-19: Use the Smith chart to find the values of Γ corresponding to the following normalized load impedances: (a) $z_L = 2 + j0$, (b) $z_L = 1 - j1$, (c) $z_L = 0.5 - j2$, (d) $z_L = -j3$, (e) $z_L = 0$ (short circuit), (f) $z_L = \infty$ (open circuit), and (g) $z_L = 1$ (matched load).

Answer: (a) $\Gamma = 0.33$, (b) $\Gamma = 0.45\angle{-63.4°}$, (c) $\Gamma = 0.83\angle{-50.9°}$, (d) $\Gamma = 1\angle{-36.9°}$, (e) $\Gamma = -1$, (f) $\Gamma = 1$, (g) $\Gamma = 0$. (See EM.)



非图2-26 P点表示归一化负载阻抗$z_L = 2 - j1$。反射系数的大小为 $|\Gamma| = OP/OR = 0.45$ 一个角 $\theta_r = -26.6°$。点 R 是 $r_L = 0$ 圆（也是 $|\Gamma| = 1$ 圆）上的任意点。

$|\Gamma| = 0.45$ 是用史密斯圆图中心与 P 点（图 2-26 中指定的 OP）之间的连线长度除以 OR 得到的，OR 是史密斯圆图中心与单位圆边缘之间的连线长度（单位圆的半径对应于$|\Gamma| = 1$）。史密斯圆图的周长包含三个同心刻度。最里面的刻度标有反射角系数（以度为单位）。这是 θ_r 的比例。如图2-26所示，P点为$\theta_r = -26.6°$ (−0.46 rad)。接下来讨论其他两个尺度的含义和用途。

练习 2-19：使用史密斯圆图找出与以下标准化负载阻抗相对应的 Γ 值：(a) $z_L = 2 + j0$，(b) $z_L = 1 - j1$，(c) $z_L = 0.5 - j2$，(d) $z_L = -j3$，(e) $z_L = 0$（短路），（f）$z_L = \infty$（开路），（g）$z_L = 1$（匹配负载）。

答案：(a) $\Gamma = 0.33$，(b) $\Gamma = 0.45\angle{-63.4°}$，(c) $\Gamma = 0.83\angle{-50.9°}$，(d) $\Gamma = 1\angle{-36.9°}$，(e) $\Gamma = -1$，(f) $\Gamma = 1$，(g) $\Gamma = 0$。（参见 EM。）

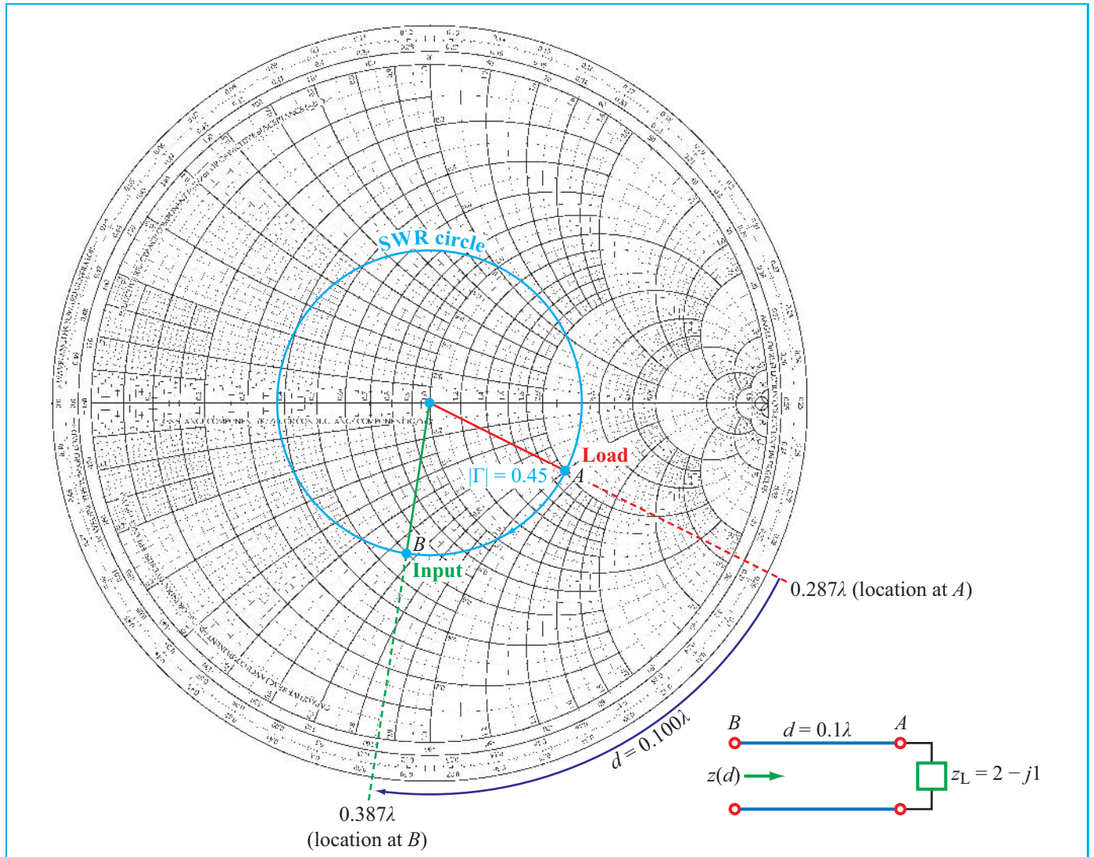


Figure 2-27 Point *A* represents a normalized load $z_L = 2 - j1$ at 0.287λ on the WTG scale. Point *B* represents the line input at $d = 0.1\lambda$ from the load. At *B*, $z(d) = 0.6 - j0.66$.

2-10.2 SWR Circle

Consider point *A* in the Smith chart of **Fig. 2-27**. At point *A*, the normalized load impedance is $z_L = 2 - j1$. The magnitude of the corresponding reflection coefficient is

$$|\Gamma| = \left|\frac{z_L - 1}{z_L + 1}\right| = \left|\frac{2 - j1 - 1}{2 - j1 + 1}\right| = \left|\frac{1 - j1}{3 - j1}\right| = \frac{\sqrt{2}}{\sqrt{10}} = 0.45.$$

Let us construct a circle centered at $(\Gamma_r, \Gamma_i) = (0, 0)$ and passing through point *A*. Every point on this circle has the same value for $|\Gamma|$, namely 0.45. This ***constant-$|\Gamma|$ circle*** is also a ***constant-SWR circle***. This follows from the relation between the voltage standing-wave ratio (SWR) and $|\Gamma|$, namely

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{2.119}$$

► A constant value of $|\Gamma|$ corresponds to a constant value of *S*, and vice versa. ◄

The utility of the SWR circle will become evident shortly.

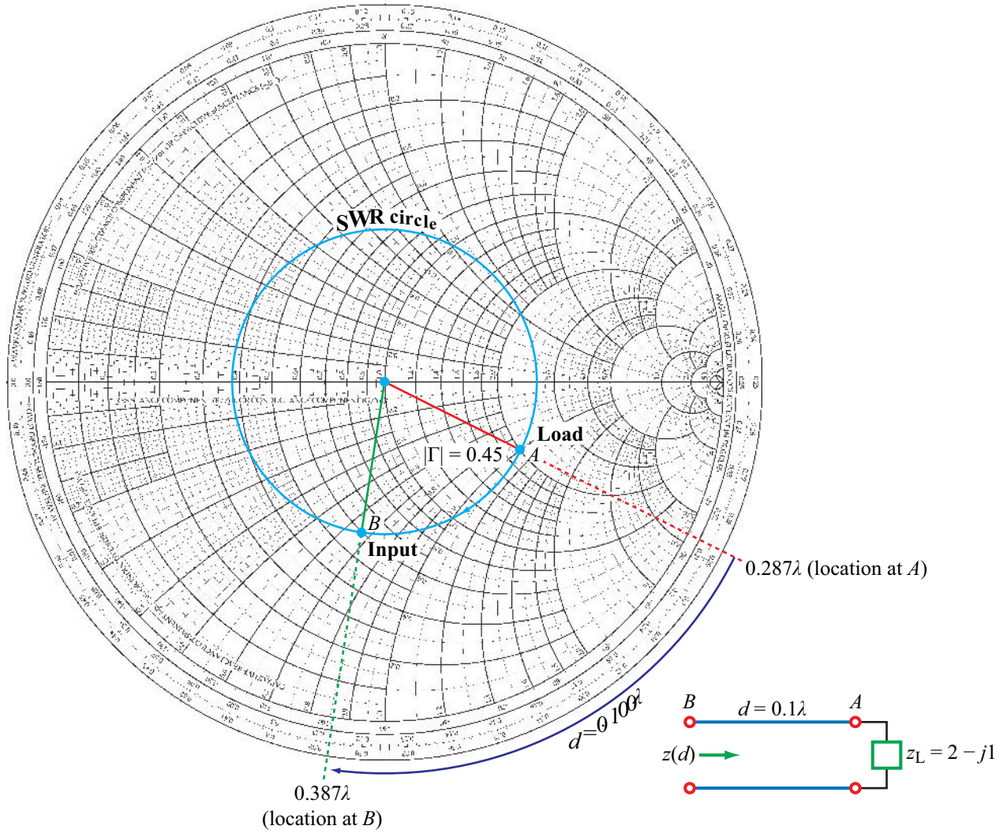


图 2-27 点 A 表示 WTG 规模上 0.287λ 处的归一化载荷 $z_L = 2 - j1$。点 B 表示来自负载的 d = 0.1λ 处的线路输入。在 B 处，z(d) = 0.6 − j0.66。

2-10.2 驻波比圆

考虑图 2-27 史密斯圆图中的 A 点。在 A 点，归一化负载阻抗为 $z_L = 2 - j1$。相应的反射系数的大小为

$$|\Gamma| = \left|\frac{z_L - 1}{z_L + 1}\right| = \left|\frac{2 - j1 - 1}{2 - j1 + 1}\right| = \left|\frac{1 - j1}{3 - j1}\right| = \frac{\sqrt{2}}{\sqrt{10}} = 0.45.$$

让我们构造一个以 $(\Gamma_r, \Gamma_i) = (0, 0)$ 为圆心并经过点 A 的圆。该圆上的每个点都有相同的 $|\Gamma|$ 值，即 0.45。这个常数-$|\Gamma|$圆也是恒定驻波比圆。这是根据电压驻波比（SWR）与$|\Gamma|$之间的关系得出的，即

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{2.119}$$

► $|\Gamma|$的常数值对应于S的常数值，反之亦然。 ◄

SWR 环的效用很快就会显现出来。

2-10.3 Wave Impedance

From Eq. (2.74), the ***normalized wave impedance*** looking toward the load at a distance d from the load is

$$z(d) = \frac{Z(d)}{Z_0} = \frac{1+\Gamma_d}{1-\Gamma_d}, \tag{2.120}$$

where

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma| e^{j(\theta_r - 2\beta d)} \tag{2.121}$$

is the *phase-shifted* voltage reflection coefficient. The form of Eq. (2.120) is identical with that for z_L given by Eq. (2.112):

$$z_L = \frac{1+\Gamma}{1-\Gamma}. \tag{2.122}$$

This similarity in form suggests that, if Γ is transformed into Γ_d, z_L gets transformed into $z(d)$. On the Smith chart, the transformation from Γ to Γ_d is achieved by maintaining $|\Gamma|$ constant and *decreasing its phase* θ_r by $2\beta d$, which corresponds to a *clockwise rotation* (on the Smith chart) over an angle of $2\beta d$ radians. A complete rotation around the Smith chart corresponds to a phase change of 2π in Γ. The length d corresponding to this phase change satisfies

$$2\beta d = 2\,\frac{2\pi}{\lambda}\,d = 2\pi, \tag{2.123}$$

from which it follows that $d = \lambda/2$.

► The outermost scale around the perimeter of the Smith chart (**Fig. 2-26**), called the ***wavelengths toward generator*** (***WTG***) scale, has been constructed to denote movement on the transmission line toward the generator, in units of the wavelength λ. That is, d is measured in wavelengths, and one complete rotation corresponds to $d = \lambda/2$. ◄

In some transmission-line problems, it may be necessary to move from some point on the transmission line toward a point closer to the load, in which case the phase of Γ must be increased, which corresponds to rotation in the counterclockwise direction. For convenience, the Smith chart contains a third scale around its perimeter (in between the θ_r scale and the WTG scale) for accommodating such an operation. It is called the ***wavelengths toward load*** (***WTL***) scale.

Exercise 2-20: Use the Smith chart to find the normalized input impedance of a lossless line of length l terminated in a normalized load impedance z_L for each of the following combinations: (a) $l = 0.25\lambda$, $z_L = 1 + j0$; (b) $l = 0.5\lambda$, $z_L = 1 + j1$; (c) $l = 0.3\lambda$, $z_L = 1 - j1$; (d) $l = 1.2\lambda$, $z_L = 0.5 - j0.5$; (e) $l = 0.1\lambda$, $z_L = 0$ (short circuit); (f) $l = 0.4\lambda$, $z_L = j3$; and (g) $l = 0.2\lambda$, $z_L = \infty$ (open circuit).

Illustration:
Using the Smith Chart to Find $Z(d)$

To illustrate how the Smith chart is used to find $Z(d)$, consider a 50 Ω lossless transmission line terminated in a load impedance $Z_L = (100 - j50)$ Ω. Our objective is to find $Z(d)$ at a distance $d = 0.1\lambda$ from the load.

1. The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{100 - j50}{50} = 2 - j1 \qquad \text{(point } A \text{ in Fig. 2-27)}.$$

2. Point A is located at 0.287λ on the WTG scale.

3. Using a compass, we construct the SWR circle with its center at the center of the Smith chart and radius passing through A.

4. As was stated earlier, to transform z_L to $z(d)$, we need to maintain $|\Gamma|$ constant, which means staying on the SWR circle, while decreasing the phase of Γ by $2\beta d$ radians. This is equivalent to moving a distance $d = 0.1\lambda$ toward the generator on the WTG scale. Since point A is located at 0.287λ on the WTG scale, $z(d)$ is found by moving to location $0.287\lambda + 0.1\lambda = 0.387\lambda$ on the WTG scale. A radial line through this new position on the WTG scale intersects the SWR circle at point B.

5. Point B represents $z(d)$, whose value is

$$z(d) = 0.6 - j0.66.$$

To obtain $Z(d)$, we unnormalize $z(d)$ by multiplying it by $Z_0 = 50$ Ω:

$$Z(d) = (0.6 - j0.66) \times 50 = (30 - j33)\ \Omega.$$

This result can be verified analytically using Eq. (2.120). The points between points A and B on the SWR circle represent different locations along the transmission line.

If a line is of length l, its ***input impedance*** is $Z_{in} = Z_0\, z(l)$ with $z(l)$ determined by rotating a distance l from the load along the WTG scale.

Answer: (a) $z_{in} = 1 + j0$, (b) $z_{in} = 1 + j1$, (c) $z_{in} = 0.76 + j0.84$, (d) $z_{in} = 0.59 + j0.66$, (e) $z_{in} = 0 + j0.73$, (f) $z_{in} = 0 + j0.72$, (g) $z_{in} = 0 - j0.32$. (See EM.)

2-10.3 波阻抗

从方程。 (2.74)，在距负载距离 d 处看向负载的归一化波阻抗为

$$z(d) = \frac{Z(d)}{Z_0} = \frac{1+\Gamma_d}{1-\Gamma_d}, \tag{2.120}$$

在哪里

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma| e^{j(\theta_r - 2\beta d)} \tag{2.121}$$

是相移电压反射系数。方程的形式(2.120) 与式 (2.120) 给出的 z_L 相同。 （2.112）：

$$z_L = \frac{1+\Gamma}{1-\Gamma}. \tag{2.122}$$

这种形式上的相似性表明，如果 Γ 转换为 Γ_d，则 z_L 也会转换为 z(d)。在史密斯圆图上，从 Γ 到 Γ_d 的变换是通过保持 |Γ| 恒定并将其相位 θ_r 减少 2β d 来实现的，这对应于顺时针旋转（在史密斯圆图上）2β d 弧度的角度。围绕史密斯圆图的完整旋转对应于 Γ 中 2π 的相变。对应于该相变的长度 d 满足

$$2\beta d = 2\,\frac{2\pi}{\lambda}\,d = 2\pi, \tag{2.123}$$

由此得出 $d = \lambda/2$。

► 史密斯圆图周边的最外层刻度（图 2-26）称为朝向发生器的波长 (WTG) 刻度，用于表示传输线上朝向发生器的移动，以波长 λ 为单位。也就是说，d 以波长为单位进行测量，完整旋转一圈对应于 $d = \lambda/2$。◄

在某些传输线问题中，可能需要从传输线上的某个点向更靠近负载的点移动，在这种情况下，必须增加 Γ 的相位，这对应于逆时针方向的旋转。为了方便起见，史密斯圆图在其周边包含第三个标度（在 θ_r 标度和 WTG 标度之间）以适应此类操作。它称为负载波长 (WTL) 标度。

练习 2-20：使用史密斯圆图计算以下每种组合的长度为 l 且终止于归一化负载阻抗 z_L 的无损线路的归一化输入阻抗： (a) $l = 0.25\lambda$， $z_L = 1 + j0$； (b)$l=0.5\lambda$，$z_L = 1+j1$； (c)$l=0.3\lambda$，$z_L = 1-j1$； (d)$l=1.2\lambda$，$z_L = 0.5-j0.5$； (e)$l=0.1\lambda$，$z_L = 0$(短路)； (f)$l=0.4\lambda$，$z_L = j3$； (g)$l=0.2\lambda$，$z_L = \infty$ (开路)。

插图：使用史密斯圆图查找 Z(d)

为了说明如何使用史密斯圆图来查找 Z(d)，请考虑终止于负载阻抗 $Z_L = (100 - j50)$ Ω 的 50 Ω 无损传输线。我们的目标是在距负载距离 $d = 0.1\lambda$ 处找到 Z(d)。

1. 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{100 - j50}{50} = 2 - j1 \qquad \text{(point } A \text{ in Fig. 2-27)}.$$

2. A点位于WTG标度上的0.287λ处。

3. 使用圆规，以史密斯圆图的中心为圆心，以 A 为半径构建 SWR 圆。

4. 如前所述，要将 z_L 转换为 z(d)，我们需要保持 |Γ| 恒定，这意味着保持在 SWR 圆上，同时将 Γ 的相位减小 2β d 弧度。这相当于在 WTG 尺度上向发电机移动距离 $d = 0.1\lambda$。由于点 A 位于 WTG 尺度上的 0.287λ，因此通过移动到 WTG 尺度上的位置 $0.287\lambda + 0.1\lambda = 0.387\lambda$ 即可找到 z(d)。通过 WTG 标尺上这个新位置的径向线与 SWR 圆相交于 B 点。

5、B点代表z(d)，其值为

$$z(d) = 0.6 - j0.66.$$

为了获得 Z(d)，我们将 z(d) 乘以 $Z_0 = 50$ Ω 来对其进行非标准化：

$$Z(d) = (0.6 - j0.66) \times 50 = (30 - j33)\ \Omega.$$

这个结果可以使用方程进行分析验证。 （2.120）。SWR 圆上 A 点和 B 点之间的点代表传输线上的不同位置。

如果线路长度为 l，则其输入阻抗为 $Z_{in} = Z_0\, z(l)$，其中 z(l) 是通过沿 WTG 标度从负载旋转距离 l 来确定的。

答案: (a) $z_{in} = 1 + j0$, (b) $z_{in} = 1 + j1$, (c) $z_{in} = 0.76 + j0.84$, (d) $z_{in} = 0.59 + j0.66$, (e) $z_{in} = 0 + j0.73$, (f) $z_{in} = 0 + j0.72$，(g) $z_{in} = 0 - j0.32$。 （参见EM。）

2-10.4 SWR, Voltage Maxima, and Minima

Consider a load with $z_L = 2 + j1$. **Figure 2-28** shows a Smith chart with a SWR circle drawn through point A, representing z_L. The SWR circle intersects the real (Γ_r) axis at two points, labeled P_{max} and P_{min}. At both points, $\Gamma_i = 0$ and $\Gamma = \Gamma_r$. Also, on the real axis, the imaginary part of the load impedance $x_L = 0$. From the definition of Γ,

$$\Gamma = \frac{z_L - 1}{z_L + 1}, \tag{2.124}$$

It follows that points P_{max} and P_{min} correspond to

$$\Gamma = \Gamma_r = \frac{r_0 - 1}{r_0 + 1} \qquad (\text{for } \Gamma_i = 0), \tag{2.125}$$

where r_0 is the value of r_L where the SWR circle intersects the Γ_r axis. Point P_{min} corresponds to $r_0 < 1$, and P_{max} corresponds to $r_0 > 1$. Rewriting Eq. (2.119) for $|\Gamma|$ in terms of S, we have

$$|\Gamma| = \frac{S - 1}{S + 1}. \tag{2.126}$$

For point P_{max}, $|\Gamma| = \Gamma_r$; hence

$$\Gamma_r = \frac{S - 1}{S + 1}. \tag{2.127}$$

The similarity in form of Eqs. (2.125) and (2.127) suggests that S equals the value of the normalized resistance r_0. By definition, $S \geq 1$, and at point P_{max}, $r_0 > 1$, which further satisfies the similarity condition. In **Fig. 2-28**, $r_0 = 2.6$ at P_{max}; hence $S = 2.6$.

> ▶ S is numerically equal to the value of r_0 at P_{max}, which is the point at which the SWR circle intersects the real Γ axis to the right of the chart's center. ◀

Points P_{min} and P_{max} also represent locations on the line where the magnitude of the voltage $|\widetilde{V}|$ is a minimum and a maximum, respectively. This is easily demonstrated by considering Eq. (2.121) for Γ_d. At point P_{max}, the total phase of Γ_d, that is, $(\theta_r - 2\beta d)$, equals zero or $-2n\pi$ (with n being a positive integer), which is the condition corresponding to $|\widetilde{V}|_{max}$, as indicated by Eq. (2.69). Similarly, at P_{min} the total phase of Γ_d equals $-(2n+1)\pi$, which is the condition for $|\widetilde{V}|_{min}$. Thus, for the transmission line represented by the SWR circle shown in **Fig. 2-28**, the distance between the load and the nearest voltage maximum is $d_{max} = 0.037\lambda$, which is obtained by moving clockwise from the load at point A to point P_{max}, and the distance to the nearest voltage minimum is $d_{min} = 0.287\lambda$, corresponding to the clockwise rotation from A to P_{min}. Since the location of $|\widetilde{V}|_{max}$ corresponds to that of $|\widetilde{I}|_{min}$ and the location of $|\widetilde{V}|_{min}$ corresponds to that of $|\widetilde{I}|_{max}$, the Smith chart provides a convenient way to determine the distances from the load to all maxima and minima on the line (recall that the standing-wave pattern has a repetition period of $\lambda/2$).

2-10.5 Impedance to Admittance Transformations

In solving certain types of transmission line problems, it is often more convenient to work with admittances than with impedances. Any impedance Z is in general a complex quantity consisting of a resistance R and a reactance X:

$$Z = R + jX \qquad (\Omega). \tag{2.128}$$

The ***admittance*** Y is the reciprocal of Z:

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2} \qquad (\text{S}). \tag{2.129}$$

The real part of Y is called the ***conductance*** G, and the imaginary part of Y is called the ***susceptance*** B. That is,

$$Y = G + jB \qquad (\text{S}). \tag{2.130}$$

Comparison of Eq. (2.130) with Eq. (2.129) reveals that

$$G = \frac{R}{R^2 + X^2} \qquad (\text{S}), \tag{2.131a}$$

$$B = \frac{-X}{R^2 + X^2} \qquad (\text{S}). \tag{2.131b}$$

A ***normalized impedance*** z is defined as the ratio of Z to the characteristic impedance of the line, Z_0. The same concept applies to the definition of the ***normalized admittance*** y; that is,

$$y = \frac{Y}{Y_0} = \frac{G}{Y_0} + j\frac{B}{Y_0} = g + jb \qquad (\text{dimensionless}), \tag{2.132}$$

where $Y_0 = 1/Z_0$ is the ***characteristic admittance of the line*** and

$$g = \frac{G}{Y_0} = GZ_0 \qquad (\text{dimensionless}), \tag{2.133a}$$

$$b = \frac{B}{Y_0} = BZ_0 \qquad (\text{dimensionless}). \tag{2.133b}$$

The lowercase quantities g and b represent the ***normalized conductance*** and ***normalized susceptance*** of y, respectively. Of course, the normalized admittance y is the reciprocal of the normalized impedance z,

$$y = \frac{Y}{Y_0} = \frac{Z_0}{Z} = \frac{1}{z}. \tag{2.134}$$

2-10.4 SWR、电压最大值和最小值

考虑 $z_L = 2 + j1$ 的负载。图 2-28 显示了史密斯圆图，其中通过点 A 绘制了 SWR 圆，表示 z_L。 SWR 圆与实轴 (Γ_r) 相交于两个点，标记为 P_{max} 和 P_{min}。在这两点上，$\Gamma_i = 0$ 和$\Gamma = \Gamma_r$。另外，在实轴上，负载阻抗 $x_L =$ 的虚部为 0。根据 Γ 的定义，

$$\Gamma = \frac{z_L - 1}{z_L + 1}, \tag{2.124}$$

由此可知点 P_{max} 和 P_{min} 对应于

$$\Gamma = \Gamma_r = \frac{r_0 - 1}{r_0 + 1} \qquad (\text{for } \Gamma_i = 0), \tag{2.125}$$

其中 r_0 是 SWR 圆与 Γ_r 轴相交处的 r_L 值。点 P_{min} 对应于 $r_0 < 1$，点 P_{max} 对应于 $r_0 > 1$。 (2.119) 对于 S 而言的 $|\Gamma|$ ，我们有

$$|\Gamma| = \frac{S - 1}{S + 1}. \tag{2.126}$$

对于点P_{max}，$|\Gamma| = \Gamma_r$；因此

$$\Gamma_r = \frac{S - 1}{S + 1}. \tag{2.127}$$

方程形式的相似性。 (2.125) 和(2.127) 表明S 等于归一化电阻r_0 的值。根据定义，$S \geq 1$，在点 P_{max} 处，$r_0 > 1$，进一步满足相似性条件。图2-28中，P_{max}处r_0 =为2.6；因此 $S = 2.6$。

> ▶ S 在数值上等于 r 的值$_0$在 P_{max}，这是 SWR 圆与真实圆相交的点Γ轴位于图表中心右侧。◀

点P_{min}和P_{max}还表示线上电压$|\widetilde{V}|$的幅度分别为最小值和最大值的位置。通过考虑等式可以很容易地证明这一点。 (2.121) 对于 Γ_d。在P_{max}点，Γ_d的总相位，即($\theta_r - 2\beta d$)等于0或$-2n\pi$ (，其中n为正整数)，这是与$|\widetilde{V}|_{max}$对应的条件，如式(1)所示。 (2.69) 。类似地，在 P_{min} 处，Γ_d 的总相位等于 $-(2n+1)\pi$，这是 $|\widetilde{V}|_{min}$ 的条件。因此，对于图2-28所示的SWR圆表示的传输线，负载与最近电压最大值之间的距离为$d_{max} = 0.037\lambda$，这是从A点的负载顺时针移动到点P_{max}得到的，而到最近电压最小值的距离为$d_{min} = 0.287\lambda$，对应于从A点顺时针旋转 A 到 P_{min}。由于 $|\widetilde{V}|_{max}$ 的位置对应于 $|\widetilde{I}|_{min}$ 的位置，而 $|\widetilde{V}|_{min}$ 的位置对应于 $|\widetilde{I}|_{max}$ 的位置，因此史密斯圆图提供了一种方便的方法来确定从负载到线路上所有最大值和最小值的距离（回想一下，驻波模式的重复周期为 $\lambda/2$）。

2-10.5 阻抗到导纳的变换 s

在解决某些类型的传输线问题时，使用导纳通常比使用阻抗更方便。一般而言，任何阻抗 Z 都是由电阻 R 和电抗 X 组成的复数：

$$Z = R + jX \qquad (\Omega). \tag{2.128}$$

导纳 Y 是 Z 的倒数：

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2} \qquad (\text{S}). \tag{2.129}$$

Y 的实部称为电导 G，Y 的虚部称为电纳 B。即：

$$Y = G + jB \qquad (\text{S}). \tag{2.130}$$

等式的比较(2.130) 与等式。 (2.129) 表明

$$G = \frac{R}{R^2 + X^2} \qquad (\text{S}), \tag{2.131a}$$

$$B = \frac{-X}{R^2 + X^2} \qquad (\text{S}). \tag{2.131b}$$

归一化阻抗 z 定义为 Z 与线路特性阻抗 Z_0 的比率。相同的概念适用于归一化导纳 y 的定义；那是，

$$y = \frac{Y}{Y_0} = \frac{G}{Y_0} + j\frac{B}{Y_0} = g + jb \qquad (\text{无量纲}), \tag{2.132}$$

其中 $Y_0 = 1/Z_0$ 是线路的特征导纳，

$$g = \frac{G}{Y_0} = GZ_0 \qquad (\text{dimensionless}), \tag{2.133a}$$

$$b = \frac{B}{Y_0} = BZ_0 \qquad (\text{dimensionless}). \tag{2.133b}$$

小写量 g 和 b 分别代表 y 的归一化电导和归一化电纳。当然，归一化导纳 y 是归一化阻抗 z 的倒数，

$$y = \frac{Y}{Y_0} = \frac{Z_0}{Z} = \frac{1}{z}. \tag{2.134}$$